Learn ಕನ್ನಡ(Kannada) interactively with Sounds By Vedavyasa Rao

By Vedavyasa Rao

Introduction

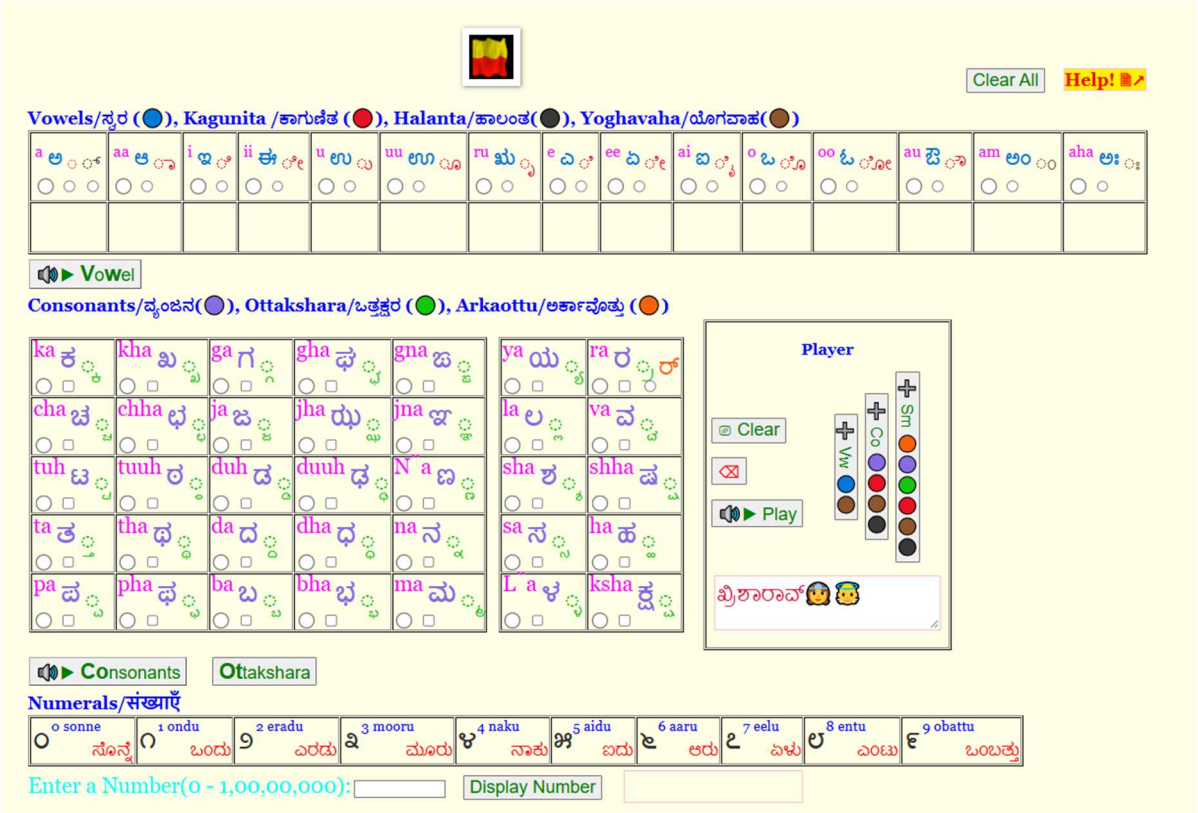
This tool is designed for the audience such as school going kids or adults interested in learning Kannada language by forming words and later hearing the sounds of the words. The sounds are provided by the courtesy of Googles Text to Speech Synthesizers.

The tool comes with preloaded sounds for vowels and consonants with Kagunita. These are again generated by the courtesy of Googles Text to Speech Synthesizers.

The preloaded sound files are stored and streamed by courtesy of GitHub.

The tool itself is a single HTML page with JavaScript code driving it. Crucial parts of the JavaScript code is generated by the courtesy of Gemini ai.

URL: https://aikannadatutor.blogspot.com/



We shall deep dive to understand different aspects of the tool.

Character Groupings

Vowels: ಅ, ಆ, ಇ, ಈ, ಉ, ಊ, ಋ, ಎ, ಏ, ಐ, ಒ, ಓ, ಔ, ಅಂ, ಅಃ

Kagunita: ◌, ◌ಾ, ◌ಿ, ◌ೀ, ◌ು, ◌ೂ, ◌ೃ, ◌ೆ, ◌ೇ, ◌ೈ, ◌ೊ, ◌ೋ, ◌ೌ

Halanta: ◌್

Yogavaha: ◌ಂ, ◌ಃ

Consonants:

ಕ, ಖ, ಗ, ಘ, ಙ,

ಚ, ಛ, ಜ, ಝ, ಞ,

ಟ, ಠ, ಡ, ಢ, ಣ,

ತ, ಥ, ದ, ಧ, ನ,

ಪ, ಫ, ಬ, ಭ, ಮ,

ಯ, ರ, ಲ, ವ, ಶ, ಷ, ಸ, ಹ, ಳ, ಕ್ಷ

Ottakshara:

◌್ಕ ◌್ಖ ◌್ಗ ◌್ಘ ◌್ಙ

◌್ಚ ◌್ಛ ◌್ಜ ◌್ಝ ◌್ಞ

◌್ಟ ◌್ಠ ◌್ಡ ◌್ಢ ◌್ಣ

◌್ತ ◌್ಥ ◌್ದ ◌್ಧ ◌್ನ

◌್ಪ ◌್ಫ ◌್ಬ ◌್ಭ ◌್ಮ

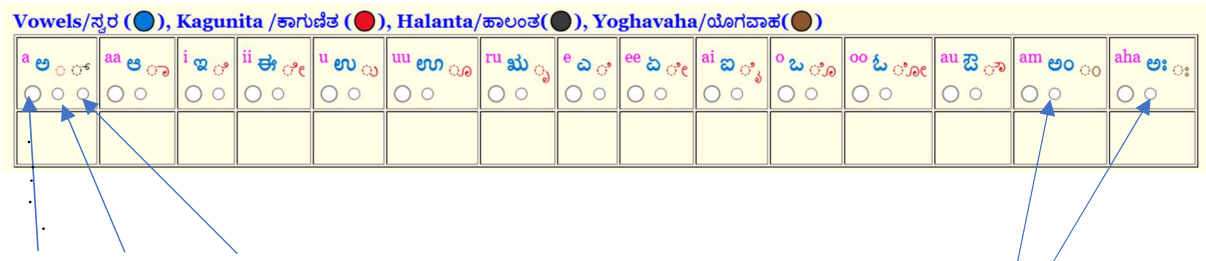
◌್ಯ ◌್ರ ◌್ಲ ◌್ವ ◌್ಶ ◌್ಷ ◌್ಸ ◌್ಹ ◌್ಳ ◌್ಕ್ಷ

Arakaouttu

ರ್

Radio button mappings

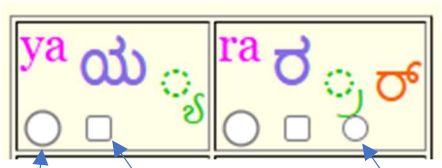
Vowels, Matras, Halant and Ayogavaha



Vowels Kagunita Halant Ayogavaha

(Anusvara, Visarga)

Consonants, Ottakshara and Arkaottu



Consonant Ottakshara Arkaottu

Notice that the radio buttons have matching colors as the text.

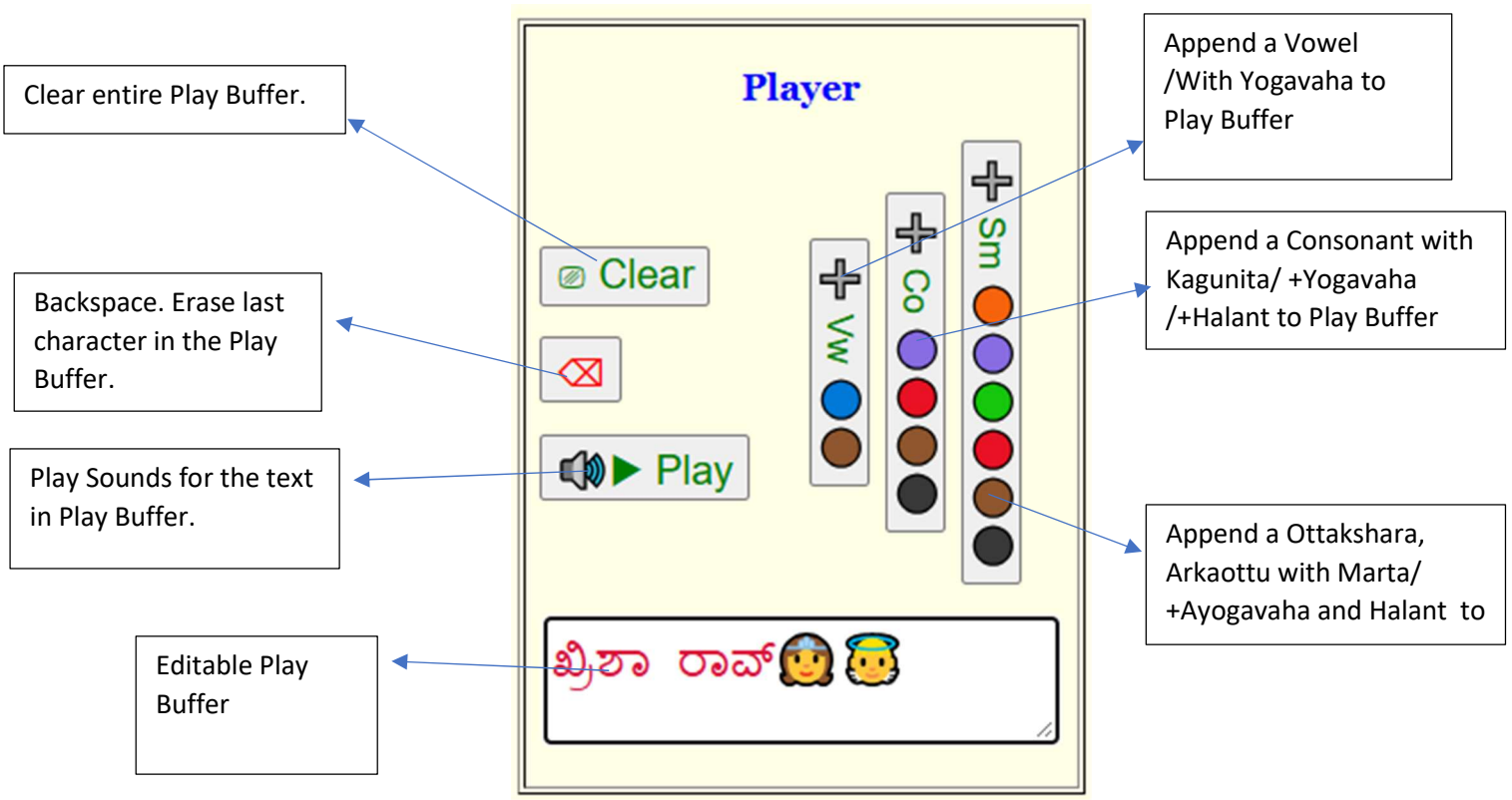


Clicking this button clears all Radio button selections.



Displays this PDF file.

Player



Grammar Rules

For Vowels, Vowels (🔵) and Yogavaha (🟤) are allowed. Examples, ಏ, ಉಂ

For Consonants, Consonants (🟣), Kagunita (🔴), Halant (⚫), Yogavaha (🟤) are allowed. Examples, ರಾಮಶಂಕರ್

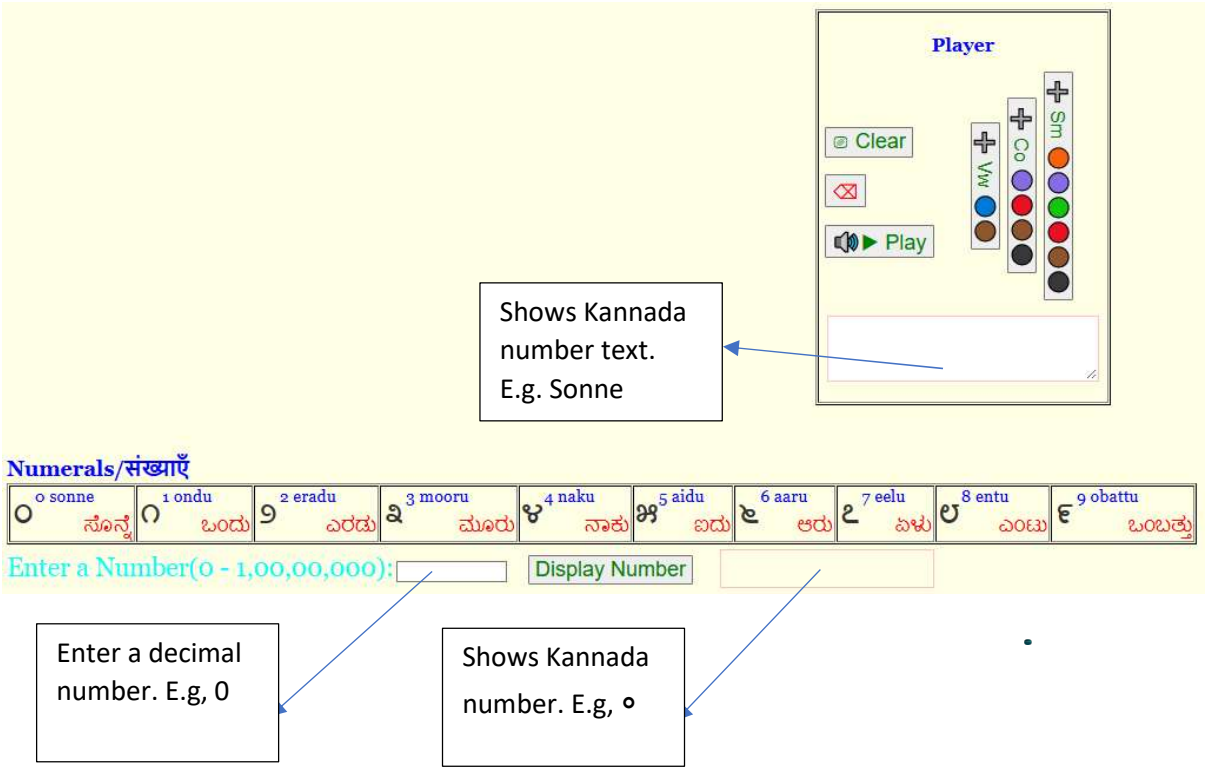
For Samyuktakshars, Ardha vyanjans (🟢), Consonants(🟣), Ayogavaha (🟤), Halant (⚫), Arkaottu (🟠) and Kagunita (🔴) are allowed. Examples, ಸೂರ್ಯಃ

The player is editable, so you can separate letters by putting comma. This will slow down the playback.

For e.g., ಕ,ಕಾ,ಕಿ,ಕೀ,ಕು,ಕೂ,ಕೃ,ಕೆ,ಕೇ,ಕೈ,ಕೊ,ಕೋ,ಕೌ,ಕಂ,ಕಃ

Numerals

The numbers can be entered into the input box. Clicking the “Display Button” prints the number and populates its Kannada text in the “Play Buffer” for audio sound as shown below.



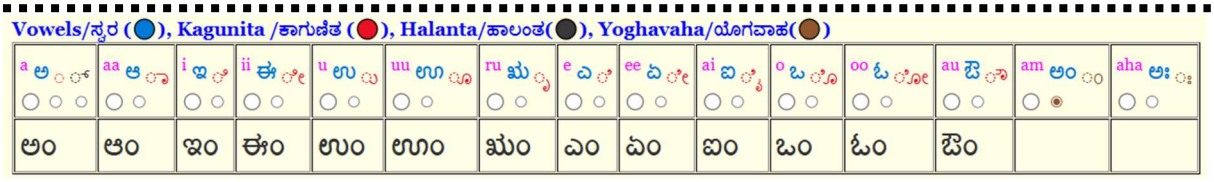
Use cases

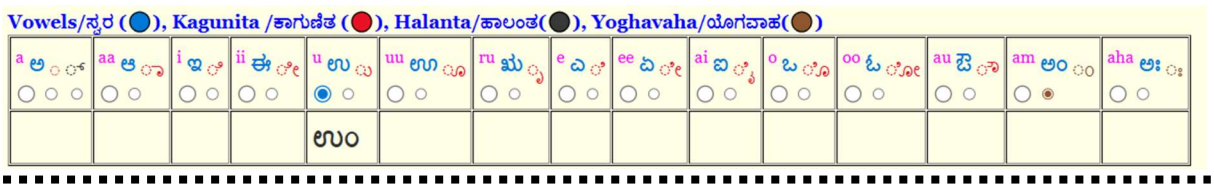


Prints Vowels and Plays associated sound.

- [x] Vowels (🔵)
- [ ] Consonants(🟣)
- [ ] Ottakshara (🟢)
- [ ] Arkaottu (🟠)
- [x] Yogavaha (🟤)
- [ ] Kagunita (🔴)
- [ ] Halant (⚫)



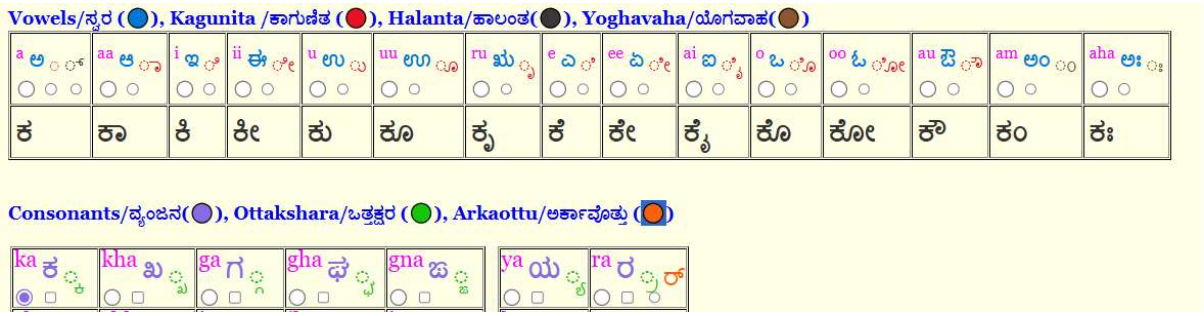


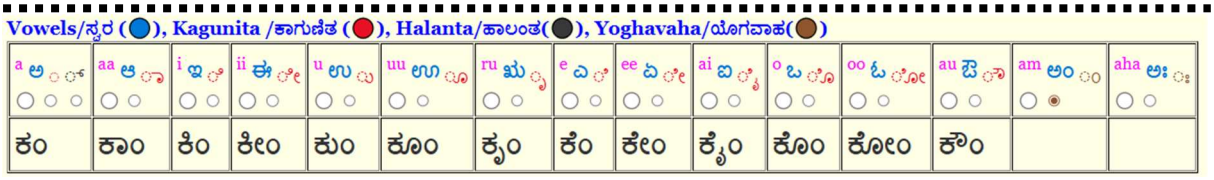


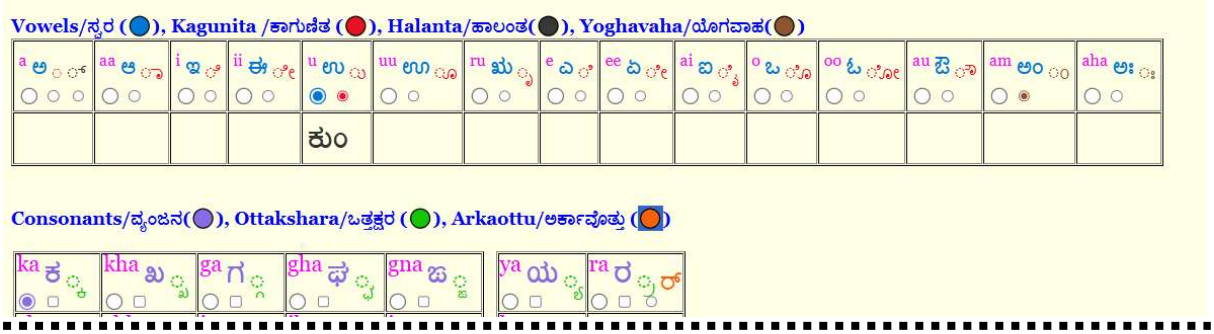


Prints Consonants and Plays associated sound.

- ✕ Vowels (🔵)
- ☑ Consonants(🟣)
- ✕ Ottakshara (🟢)
- ✕ Arkaottu (🟠)
- ☑ Yogavaha (🟤)
- ☑ Kagunita (🔴)
- ☑ Halant (⚫)







Samyuktakshars

Prints Samyuktakshars only. The sounds can be generated from the player by separating generated text with a comma as shown below.

For e.g., क्र,क्रा,क्रि,क्री,क्रु,क्रू,क्रृ,क्रे,क्रै,क्रो,क्रौ,क्रं,क्रः

✕ Vowels (🔵)

☑ Consonants(🟣)

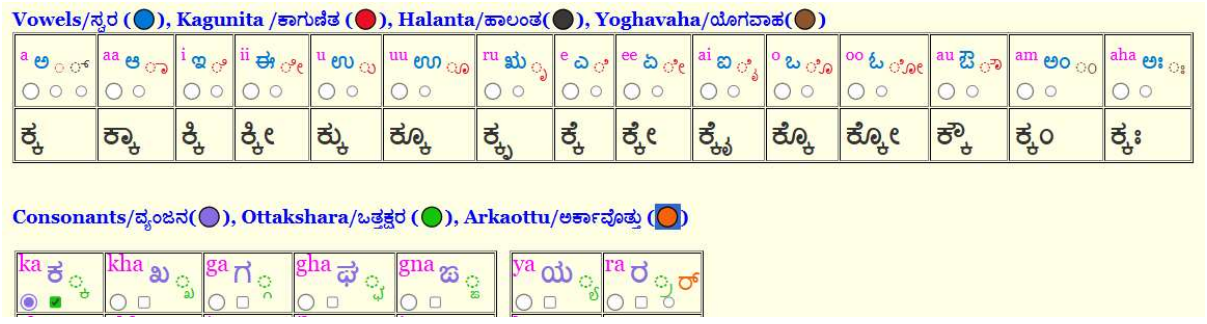
☑ Ottakshara (🟢)

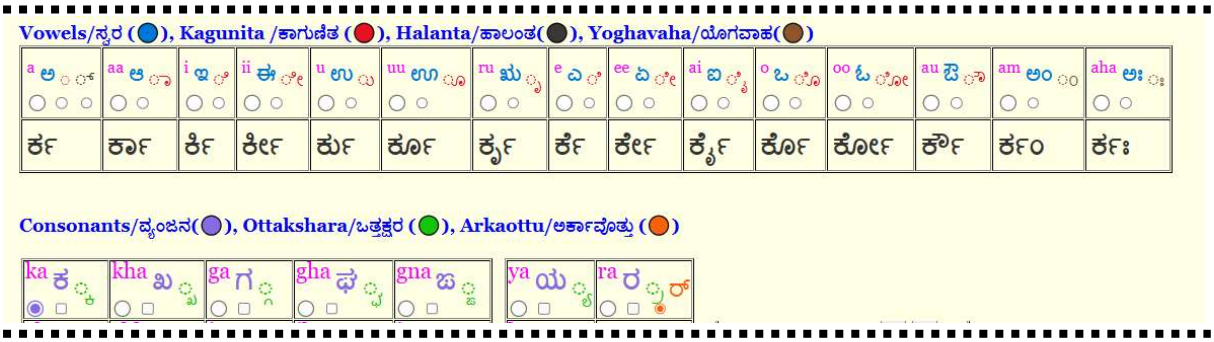
☑ Arkaottu (🟠)

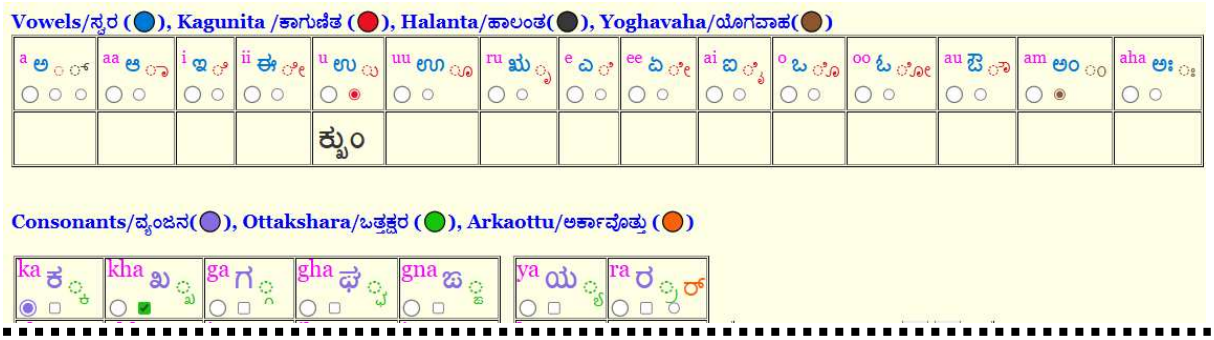
☑ Yogavaha (🟤)

☑ Kagunita (🔴)

☑ Halant (⚫)





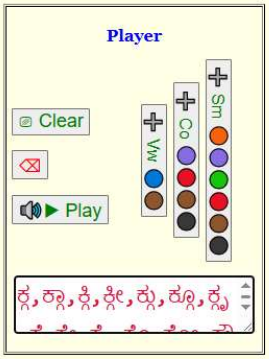


Vowels/ಸ್ವರ (🔵), Kagunita /ಕಾಗುಣಿತ (🔴), Halanta/ಹಾಲಂತ(⚫), Yoghavaha/ಯೊಗವಾಹ(🟤)

a ಅ ◌ ◌್	aa ಆ ◌ಾ	i ಇ ◌ಿ	ii ಈ ◌ೀ	u ಉ ◌ು	uu ಊ ◌ೂ	ru ಋ ◌ೃ	e ಎ ◌ೆ	ee ಏ ◌ೇ	ai ಐ ◌ೈ	o ಒ ◌ೊ	oo ಓ ◌ೋ	au ಔ ◌ೌ	am ಅಂ ◌ಂ	aha ಅಃ ◌ಃ
○ ○ ○	○ ○	○ ○	○ ○	○ ○	○ ○	○ ○	○ ○	○ ○	○ ○	○ ○	○ ○	○ ○	○ ○	○ ○
ಕ್ಷ	ಕ್ಷಾ	ಕ್ಷಿ	ಕ್ಷೀ	ಕ್ಷು	ಕ್ಷೂ	ಕ್ಷೃ	ಕ್ಷೆ	ಕ್ಷೇ	ಕ್ಷೈ	ಕ್ಷೊ	ಕ್ಷೋ	ಕ್ಷೌ	ಕ್ಷಂ	ಕ್ಷಃ

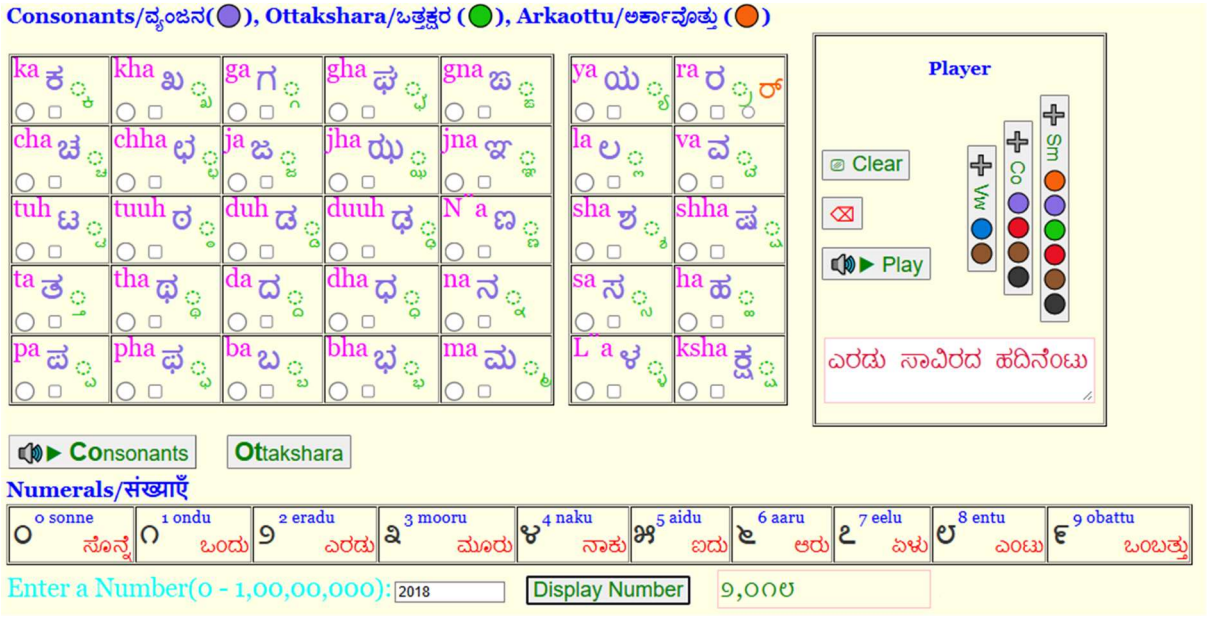
Consonants/ವ್ಯಂಜನ(🟣), Ottakshara/ಒತ್ತಕ್ಷರ (🟢), Arkaottu/ಅರ್ಕಾವೊತ್ತು (🟠)

ka ಕ ◌್ಕ	kha ಖ ◌್ಖ	ga ಗ ◌್ಗ	gha ಘ ◌್ಘ	gna ಙ ◌್ಙ	ya ಯ ◌್ಯ	ra ರ ◌್ರ ◌ರ್
◉ □	○ □	○ ☑	○ □	○ □	○ □	○ □
cha ಚ ◌್ಚ	chha ಛ ◌್ಛ	ja ಜ ◌್ಜ	jha ಝ ◌್ಝ	jna ಞ ◌್ಞ	la ಲ ◌್ಲ	va ವ ◌್ವ
○ □	○ □	○ □	○ □	○ □	○ □	○ □
tuh ಟ ◌್ಟ	tuuh ಠ ◌್ಠ	duh ಡ ◌್ಡ	duuh ಢ ◌್ಢ	N a ಣ ◌್ಣ	sha ಶ ◌್ಶ	shha ಷ ◌್ಷ
○ □	○ □	○ □	○ □	○ □	○ □	○ □
ta ತ ◌್ತ	tha ಥ ◌್ಥ	da ದ ◌್ದ	dha ಧ ◌್ಧ	na ನ ◌್ನ	sa ಸ ◌್ಸ	ha ಹ ◌್ಹ
○ □	○ □	○ □	○ □	○ □	○ □	○ □
pa ಪ ◌್ಪ	pha ಫ ◌್ಫ	ba ಬ ◌್ಬ	bha ಭ ◌್ಭ	ma ಮ ◌್ಮ	L a ಳ ◌್ಳ	ksha ಕ್ಷ ◌್ಕ್ಷ
○ □	○ □	○ □	○ □	○ □	○ □	○ □



Display Number

Converts decimal number into Kannada number and kannada text for the same.



Troubleshooting and workarounds

Preloaded Sounds

Sometimes the Audio files don't get loaded for streaming. In such cases, the Vowel or Consonants button background color becomes red as shown below.



First make sure radio button selections are made to play sounds.

You can retry by clicking on the red button several times till it becomes green when reloading becoming successful.

If it still stays red, reload the page in the browser.

If it still stays red, you can continue and use player for sounds.

Sound Quality on Mobile devices

Due to CORS network restrictions, locally installed device voices are used instead of high-quality Google neural models. For the best sound quality and experience, please use a Desktop or Laptop if possible.

Reach out!

If you observe any inaccuracies or inconsistencies, kindly drop me an email at rvvyas@gmail.com with the subject starting with "aikannadatutor".